

Evaluating Adaptive Feedback Mechanisms in AI Tutoring Systems

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ABSTRACT

This empirical investigation assesses the efficacy of generative AI-powered tutoring systems on knowledge acquisition and cognitive retention. In a randomized controlled experiment involving 450 undergraduate engineering students across two semesters, participants receiving AI adaptive feedback demonstrated a 24.8% improvement in retention over passive lecture baselines ($p < 0.01$). However, self-efficacy gains were moderated by baseline domain familiarity, revealing that novice learners experienced heightened initial confusion.

1. INTRODUCTION

Adaptive tutoring has emerged as a cornerstone of modern educational computing. Recent breakthroughs in transformer architectures enable conversational pedagogical agents to provide granular, step-by-step guidance. Despite rapid deployment, rigorous empirical validation of learning longevity remains sparse across multi-cohort samples.

2. METHODOLOGY AND EXPERIMENTAL SETUP

We executed a double-blind trial with 450 undergraduate participants enrolled in introductory STEM sequences. The intervention group ($n=225$) utilized our adaptive LLM scaffolding module, whereas the control group ($n=225$) utilized standard digital problem sets. Pre-tests and post-tests measured conceptual transfer across three distinct difficulty tiers.

3. EMPIRICAL RESULTS AND FINDINGS

Quantitative test outcomes revealed significant performance variance. Participants interacting with real-time adaptive feedback demonstrated statistically significant improvements on problem-solving speed ($t=4.12$, $p<0.001$) and complex concept transfer. Specifically, formative assessment scores improved by an average of 24.8% relative to control groups.

4. LIMITATIONS

Several constraints contextualize our findings. First, the trial was conducted solely within single-institution engineering courses, potentially biasing findings toward mathematically inclined cohorts. Second, longitudinal retention beyond a 16-week observation period was not tracked. Third, qualitative interviews indicated minor cognitive fatigue during prolonged interactive sessions.

5. FUTURE WORK AND CONCLUSIONS

Future investigations must deploy multi-institutional longitudinal benchmarks across non-STEM disciplines and investigate personalized pacing mechanisms to alleviate learner fatigue. In conclusion, AI tutoring provides substantial empirical advantages when anchored by structured pedagogical frameworks.